

Disproof of Conjecture 00000003940

earthking11

October 6, 2026

The incompatibility

The conjecture defines a finite-type invariant as a numerical invariant obtained from a finite truncation of a differential graded algebra (DGA). It then asks for two Legendrians whose DGAs are chain-isomorphic while all their finite-type invariants differ.

Let $\Phi : A \rightarrow B$ be a chain-level DGA isomorphism. Because Φ preserves the differential, multiplication, grading, and filtration, for every truncation level n it induces an isomorphism

$$\Phi_{\leq n} : A_{\leq n} \xrightarrow{\sim} B_{\leq n}.$$

If I_n is a numerical invariant obtained from the n th truncation, invariance under isomorphism gives

$$I_n(A) = I_n(B).$$

This holds for every n .

Proposition 1. *No two chain-isomorphic DGAs can have all truncation invariants different.*

Proof. For an arbitrary truncation level n , restrict the chain isomorphism to that truncation. Since I_n is an invariant, the displayed equality follows. It contradicts the assertion $I_n(A) \neq I_n(B)$, already for one n and hence a fortiori for all n . $\square$

The crossing-number condition “at most 12” is irrelevant: the requested pair cannot exist at any crossing number. Calling a numerical construction an “invariant” precisely commits it to being constant on the relevant isomorphism classes.

The accompanying Lean project formalizes this general implication and a concrete list-truncation model.